Lantern Office Suite: v1.6 Review and Rebuild

October 19, 2026 · Repo: **github.com/adamweldon/Lantern_Office_Suite** (private) · Build 1.6.0

188	41	10/10	0
modules (was 160)	apps, all on first paint	test checks pass	errors in real Chromium

> **Bottom line.** Your v1_5 upload was the July 27 file from *before* the menu fix. Since then four sessions of work were missing. I rebuilt everything whose source survives in our chats, fixed the first-paint bug, and applied the corrections those chats flagged. The result is in the repo as `Lantern_Office_Suite.html` (1.1 MB, one file, offline). Two pieces could not be recovered; see "Still missing".

What happened to v1.6

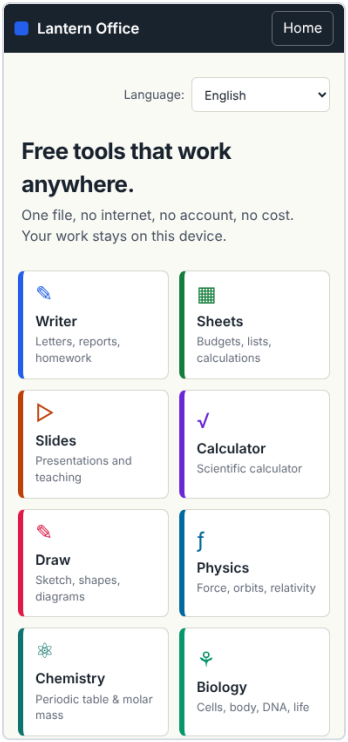
No chat ever built v1.6. It only appears as your upload in the August 18 iPhone chat: 252 modules plus a separate Studio (Generative Music Studio) script block. You assembled it yourself in Project Context Studio. Chat transcripts keep tool *inputs* but not tool *outputs*, so the v1.6 module list itself is gone.

The inputs are enough, though. They hold every file Claude wrote and every edit command it ran. I re-executed those sessions in a sandbox, so their own code regenerated their own deliverables, with nothing retyped.

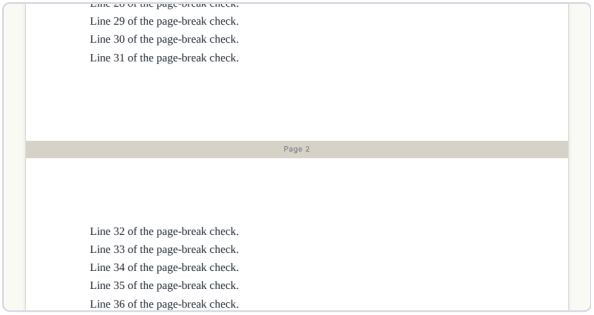
- **Calibration.** Replaying the July 15 RF Bench session regenerated its 11 modules **byte-identical** to the copy in your file (75,965 characters). That proved the method before I trusted it on anything new.
- **Faithful to the originals.** Each replayed session's own test suites were re-run and pass, on its output and on the final combined build.

What was restored

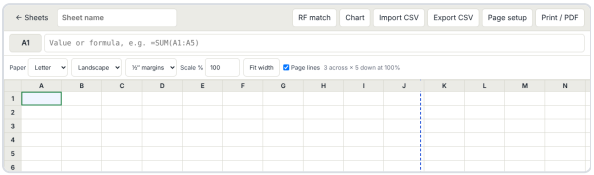
Work	From	Modules	Proof
FIX BOOT first paint: only 24 of 37 apps appeared until you navigated away and back	Jul 27	BOOT	Real Chromium: baseline 24/37, rebuild 41/41
ADD Health split: staying-well vs emergencies, 6 new lessons (fever, teeth, breaks, infection, heat/cold, bites)	Jul 13	HEALTH_LESSONS_EMERGENCY + 3 updated	All 8 old lessons kept; 157 other modules byte-identical
ADD Paper sizes, real page breaks, page setup and guides; rule 11 (isolation) and hooks; Writer word-count fix	Jul 13	WRITER_PAGES, SHEET_PAGES, SHEET_PRINT, SHEET_CSV, GUIDE_PAGES	That session's 26-check smoke and geometry suites pass
ADD Deckplate music player	Jul 20	10 DECKPLATE_*	Hidden/display regressions and suite smoke pass
ADD AI tool layer: Astro cells, Assist, Forge, Chat (proposals staged, never auto-applied)	Jul 20	ASTRO_CELLS, ASSIST_*, MOD_FORGE, FORGE_APP, ASSIST_CHAT, GUIDE_AI	Assist/Astro ground truth and Forge gate/rollback suites pass



Home on a 390-px phone screen.



Writer, Letter paper: text continues at the top of page 2.



Sheets: RF match, CSV, Page setup and Print in one toolbar; dashed page guides.

Corrected

- **Quotes inside regex literals.** There were 6 in CORE (escaping and CSV), WRITER_IMPORT (sanitizer) and ASSIST_CORE. The July 27 audit found these desync Project Context Studio's gate, which passed them only by luck. Each is now written as `\x22` / `\x27`, and a 4,000-string fuzz test confirms identical matching.
- **Size limit.** REPAIR_LESSONS (1,771 words) and LOCALE_WORLD (1,896) were split at nameable boundaries: bicycle vs small engines, and Europe/Africa vs Asia packs. Every registration statement is unchanged and in order; the runtime still has 252 lessons and 9 languages.

Still missing

Item	Why	To restore
GONE Flight, Build and Wilderness apps; 16 cell functions (=NACA, =LIFT, =REYNOLDS, =MACH, =BOARDFEET…); tiered physics lessons; PHYSICS_DRAWINGS; the illustration pass	The July 7 chat is withheld (safety-classifier redaction); only search fragments are readable	Upload any file that has them, such as July 7's `lantern-office.html` v1.6.1. Its Repair app is a different design from yours (fault-finding vs bicycle/engine), so it needs a merge decision.
GONE Studio (Generative Music Studio) tile	Merged outside our chats	Upload your current GMS file and I'll add it as an isolated tile.

Open items (not changed)

- **Code modules over the size limit.** SLIDES (1,504 words), DRAW (1,715), RF_BENCH_APP (1,699) and DECKPLATE_TAGS (2,339) are working code. Splitting them is a refactor, so I left them alone.
- **iPhone, from Aug 18.** Drag interactions (column resize, page guides, Studio piano roll) don't respond to touch. Storage on `file://` may not persist in the iOS Files preview. The offered pointer-events shim and boot-time storage probe were never built.
- **BOOT sits mid-file** (rule 1). This is harmless now that the start is deferred.
- **Repacking.** Project Context Studio's merge adds one newline per module boundary per pass and does not re-bundle. Repack with `npm run build`, and run `npm test` before sharing.

Verification

- `npm test`: 10 checks.
  - The bundle decodes exactly to the source.
  - Structure: markers balanced, no code outside modules, ES5–ES2017 only, parses.
  - No quotes inside regex literals.
  - Boot tour: runs the script after DOMContentLoaded the way the bundle does, then opens every app and fails on any exception.

    - The six recovered session suites.
- **Real Chromium**, opening the shareable file over `file://` through its own unpacker at desktop and phone widths: 41/41 tiles, every app opens, 0 page errors. Writer shows pages 2 and 3 after 70 lines.
- **Module diff against your upload:** 28 added, 10 changed (each traceable to a commit), 150 byte-identical, none removed.

Commits on main: baseline (exact upload) → BOOT fix → Health → Deckplate + AI → Pages → regex quotes → size splits → tests → 1.6.0 + CHANGELOG. Each one can be reverted on its own.